Roman Ostafeichuk

Systems Programming • Compilers • C/C++

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Profile

MIPT student focused on systems programming, compilers, and low-level C/C++ development. Interested in AST/IR pipelines, interpreters, Linux/POSIX, GPU programming, and infrastructure for ML graphs.

Education

Moscow Institute of Physics and Technology, FRTC

2024 — present

Applied Mathematics and Physics

- GPA: 7.04 / 10

Additional Coursework

Systems Programming and Compiler Technologies

1st semester

Instructor: I. R. Dedinsky

MIPT ILab C++ Course

Grade for 3rd semester: excellent, 8

Instructor: K. I. Vladimirov

Systems Programming

by VK Education

Introduction to Tensor Compilers

in progress

Introduction to Computer Architecture

in progress

Skills

Programming Languages: C, C++, Python, Bash

Systems Programming: Linux, POSIX API

Tools: Git, Make, CMake, GTest, CTest, GitHub Actions

Technologies and Libraries: Flex, Bison, OpenCL, SQLite, libuv, ncurses, Protobuf, ONNX, Graphviz

Languages: Russian (native), English (A2–B1)

Selected Projects

ParaCL — team project: interpreter for a C-like language

C++20 · Flex · Bison · CMake · Graphviz

- Responsible for AST, Visitor infrastructure, and the interpreter.
- Contributed to AST visualization and project engineering setup: tests, CI, sanitizers.

TensorCompiler — team project: frontend for ONNX models

C++17 · CMake · Protobuf · ONNX · Graphviz

- Worked on importing ONNX models into an internal IR graph representation.
- Implemented model verification and worked with Graph / Node / tensor attribute structures.

Bitonic Sort on GPU

C++17 · OpenCL · Python

- Implemented bitonic sort with OpenCL kernels and a C++ host-side application.
- Added end-to-end validation, benchmarks, and CI for Linux/Windows.

SLR(1) Parser for an Arithmetic Language

C++17 · Flex · CMake

- Implemented a table-driven SLR(1) shift-reduce parser for an arithmetic language.
- Used Flex; ACTION/GOTO tables are stored in CSV and embedded during build.

Mini Command-Line Shell

C · Linux · POSIX

- Implemented a Bash-like parser and command execution engine.
- Added pipelines, output redirection, and Unix process handling.

Assembler + Virtual Machine

C/C++

- Developed an educational toolchain: an assembler and a virtual machine for binary code execution.
- Worked on instruction format, VM logic, and smoke tests.

Additional Projects

- `cache_task` — **LRU · LIRS · Ideal Cache** — comparison of caching strategies.
- `Coroutine_messaging_system` — **C++ · coroutines · channels** — cooperative coroutines and bounded channels without threads.
- `Sea_battle` — **C · libuv · SQLite · ncurses** — client-server game.
- `RB-tree` — **C++ · data structures** — red-black tree implementation with balancing and tests.
- `Differentiater` — **C++ · AST · Graphviz** — symbolic differentiation and expression visualization.